SHUBHAM PANWAR

Lead Engineer • React.js | Redux | TypeScript | Next.js

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PROFESSIONAL SUMMARY

Lead Engineer with 8 years of experience architecting and shipping production React applications across AI, PropTech, and automation products. Most recently led the front-end of an AI-powered product-costing platform, turning LLM capabilities into reliable, explainable product experiences. Deep expertise across the React ecosystem (Hooks, Redux/Redux Toolkit, Context) plus TypeScript and Next.js. Proven track record of cutting load times by 40–60%, leading component-library initiatives, mentoring engineers, and owning features end-to-end from design hand-off to production. Equally comfortable making architectural decisions, optimizing Core Web Vitals, and partnering with product, design, AI, and backend teams in early-stage and fast-paced environments.

CORE SKILLS

Languages: JavaScript (ES6+), TypeScript, HTML5, CSS3
AI / LLM: Gemini & Claude APIs, multi-step agent workflows, prompt engineering, structured outputs, retrieval-grounded validation (RAG), multimodal inputs, LLM evaluation, AI cost monitoring
Frameworks & Libraries: React.js, Redux, Redux Toolkit, Next.js, React Router, Zustand, Node.js, Express
Styling: Tailwind CSS, Styled Components, SCSS/SASS, CSS Modules
Testing & Monitoring: Vitest, React Testing Library, Playwright, Storybook, Sentry
APIs & Data: REST, WebSockets, Axios, MongoDB
Cloud & Practices: GCP (Cloud Run, Cloud Functions), Component Architecture, Performance Optimization, CI/CD, Code Review, Agile/Scrum

PROFESSIONAL EXPERIENCE

Lead Engineer | Naya Studio

March 2022 – Present

Early-stage startup building an AI-powered product cost estimation platform for enterprise clients across consumer electronics, furniture, apparel, and industrial manufacturing. naya.studio

- Lead the front-end for Naya's AI Estimation and Workflow products, React + TypeScript apps that turn sketches, 3D models, BOMs, and PDFs into manufacturing cost estimates for enterprise teams. Architected the core workflow modules: BOM/CBD ingestion, 3D previews (GLB/OBJ/STL/STEP), image-to-3D generation, and real-time cost reports.
- Architected the product experience for a multi-step LLM agent workflow that drafts a plan, then executes it step by step with shared context. It supports multimodal inputs (images, PDFs, spreadsheets, CAD), Gemini and Claude via a secure gateway, industry-specific prompt skills with AI domain detection, and parallel batch runs with AI spend tracking and usage limits.
- Designed a hybrid AI + rules approach for reliable, explainable estimates. The LLM selects rates and cost structure, and a rule-based formula engine does the arithmetic. Added web-search-grounded validation (RAG) and schema-validated structured outputs to reduce hallucination, and led prompt hardening and evaluation against real factory cost data.
- Built human-in-the-loop review flows that let users override and pin values and rerun individual agent steps, with AI-sourced values clearly badged, giving enterprise users transparency and control over every estimate.
- Reduced Largest Contentful Paint by ~55% and Time to Interactive by ~45% by introducing route-based code splitting, lazy loading, image optimization, and React.memo/useMemo audits across critical user journeys.
- Designed and rolled out an internal React component library in TypeScript + Storybook adopted across product surfaces, cutting new-feature UI build time by ~40% and enforcing a consistent design language across teams.
- Migrated a legacy Redux codebase to Redux Toolkit, eliminating ~35% of boilerplate and removing an entire class of race-condition bugs in asynchronous data flows.
- Set engineering direction for the front-end team, including hiring loops, RFC/architecture reviews, code-review standards, and performance budgets, and mentor engineers through pairing and structured 1:1s.
- Administer 50+ GitHub repositories: automated branch protection, GitHub Actions release pipelines, and semantic-release versioning.
- Drove SOC 2 readiness solo, owning security controls, access policies, and audit evidence across the engineering org and partnering with auditors through to completion.

Senior Software Engineer | Rently *Feb 2021 – Feb 2022*

Product company building a resident-lifecycle management platform with self-guided tours, smart home, and access-control solutions for property managers. use.rently.com

- Built and owned features of a React + Next.js application (SSR + ISR) powering self-guided tour scheduling and smart-home management used by leading US property operators.
- Led the migration from Create React App to Vite, reducing local dev server startup from ~18s to under 2s and cutting CI build times by ~45%.
- Implemented a real-time scheduling and access-control module using WebSockets and optimistic UI with Redux Toolkit, improving perceived responsiveness and reducing support escalations.
- Introduced React Testing Library to the stack, raising automated test coverage on core flows from ~20% to 75% and reducing production regressions quarter-over-quarter.
- Partnered with backend and product teams on REST API design and contracts, and on sprint planning in a distributed agile team.

Software Engineer | Relfor Labs *Sept 2018 – Feb 2021*

Startup building restaurant automation and point-of-sale software for QSR and dine-in restaurant chains.

- Developed responsive, cross-browser React.js interfaces for restaurant POS, order-management, and kitchen-display systems used across multiple restaurant locations — working end-to-end from wireframes to production.
- Built reusable Redux state slices and custom hooks for cart, order, and authentication flows, reducing duplicated logic across multiple product surfaces (POS, admin console, customer-facing ordering).
- Integrated payment gateways (Razorpay) and hardened error handling and retry logic, contributing to a measurable drop in failed-payment support tickets.
- Contributed to migrating parts of a legacy jQuery codebase to React, incrementally modernizing a significant portion of the application surface area.
- Worked directly with founders and product in a small, fast-moving team across 2-week sprints; consistently delivered committed scope through planning, retros, and demos.

EDUCATION

Bachelor of Engineering (B.E.) | RGPV University *2013 – 2017*

Diploma in Advanced Computing | CDAC-ACTS, Pune *Feb 2018 – July 2018*